

Tendernomics

E-Commerce Insights with Microsoft Fabric & Power BI

From Data to Decisions

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Problem Statement

As an e-commerce platform owner, I want to understand what factors impact customer satisfaction and delivery performance so I can improve overall customer experience.

What is Olist

Olist connects small businesses from all over Brazil to channels without hassle and with a single contract.



The screenshot shows the Olist website's homepage. At the top, there's a navigation bar with links for 'Products', 'Solutions', 'Large companies', 'More', and 'Plans', along with 'login' and a 'free trial' button. The main headline reads 'Fewer tools, more sales results', followed by a sub-headline: 'Olist is the ecosystem of solutions that streamlines and automates the management of Brazilian entrepreneurs'. Below this are buttons for '30-day free trial' and 'see plans'. A background image shows a receipt printer. A large quote states: 'With cutting-edge technology, entrepreneurship is easier. It's that simple.' At the bottom, a circular diagram illustrates the Olist ecosystem, with 'olist' at the center and various business functions radiating outwards: AI Assistant, Shipping, Sales channels, Store, Inventory management, Integration Hub, Advertisements, ERP, Business management, Your company, Digital Account, Ecommerce, POS System, Reports, Financial management, and Supplies.

Olist item listing snippet

- Item review rating
- Price
- Payment Options

Smartphone Motorola Moto G6 Play Dual Chip Android Oreo - 8.0 Tela 5.7" Octa-Core 1.4 GHz 32GB 4G Câmera 13MP - Índigo

(Cod:133453125) **★★★★★ (215)**

☐ Caixa de Som ANKER SoundCore Bluetooth 12W - Preta + R\$ 429,99

pegue na loja hoje! Pegue na loja mais próxima, no mesmo dia :) Sujeito à alteração de preço. [Saiba mais](#) [ver lojas](#)

Escolha uma loja abaixo e compre

R\$ 1.299,00
R\$ 26,04 - 7 a 10 dias úteis

☐ on-line
R\$ 1.069,90
R\$ 38,32 - 7 a 10 dias úteis

☐ mel
R\$ 975,00
R\$ 22,94 - 5 a 6 dias úteis

Mais opções deste produto a partir de **R\$ 959,00**

R\$ 1.299,00
10x de R\$ 129,90 s/ juros

comprar

Corra! Temos apenas 5 no estoque

R\$ 1.299,00 em até 12x de R\$ 108,25 s/ juros

R\$ 1.299,00 no cartão : em até 24x de R\$ 54,12 s/ juros

[formas de parcelamento](#)

:) Este produto é vendido por uma loja parceira.

Customer Journey

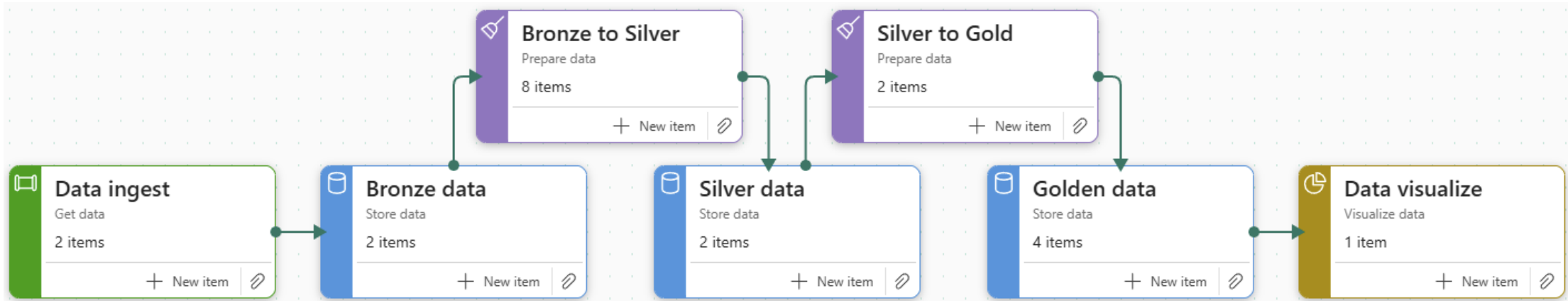
Journey Stages:

Awareness > Consideration > Fulfillment > Retention

Key Decision Factors:

- Price: Competitiveness and value perception
- Rating: Trust and previous customer experiences
- Purchase Options: Convenience, payment methods, and delivery flexibility

Solution Architecture



- Microsoft Fabric
 - Notebooks (PySpark) - Ingestion and Transformation
 - Lakehouse – Storage
- Power BI - Visualization

Datasets Overview

- 9 raw datasets ingested in our lakehouse
 - Payments, Orders, Order Items, Products, Sellers, Customers, Product Category Translation, Geolocation
- ~100K orders, 90K customers (Sept 2016 – Oct 2018)
- Automated EDA with Pandas Profiling library

Data Cleaning & Validation Highlights

1) Chronological Order Enforcement

 Purchase →  Approval →  Courier Pickup →  Delivered

- Ensured milestone timestamps followed business logic
- Inaccurate sequencing errors → quarantined for data traceability

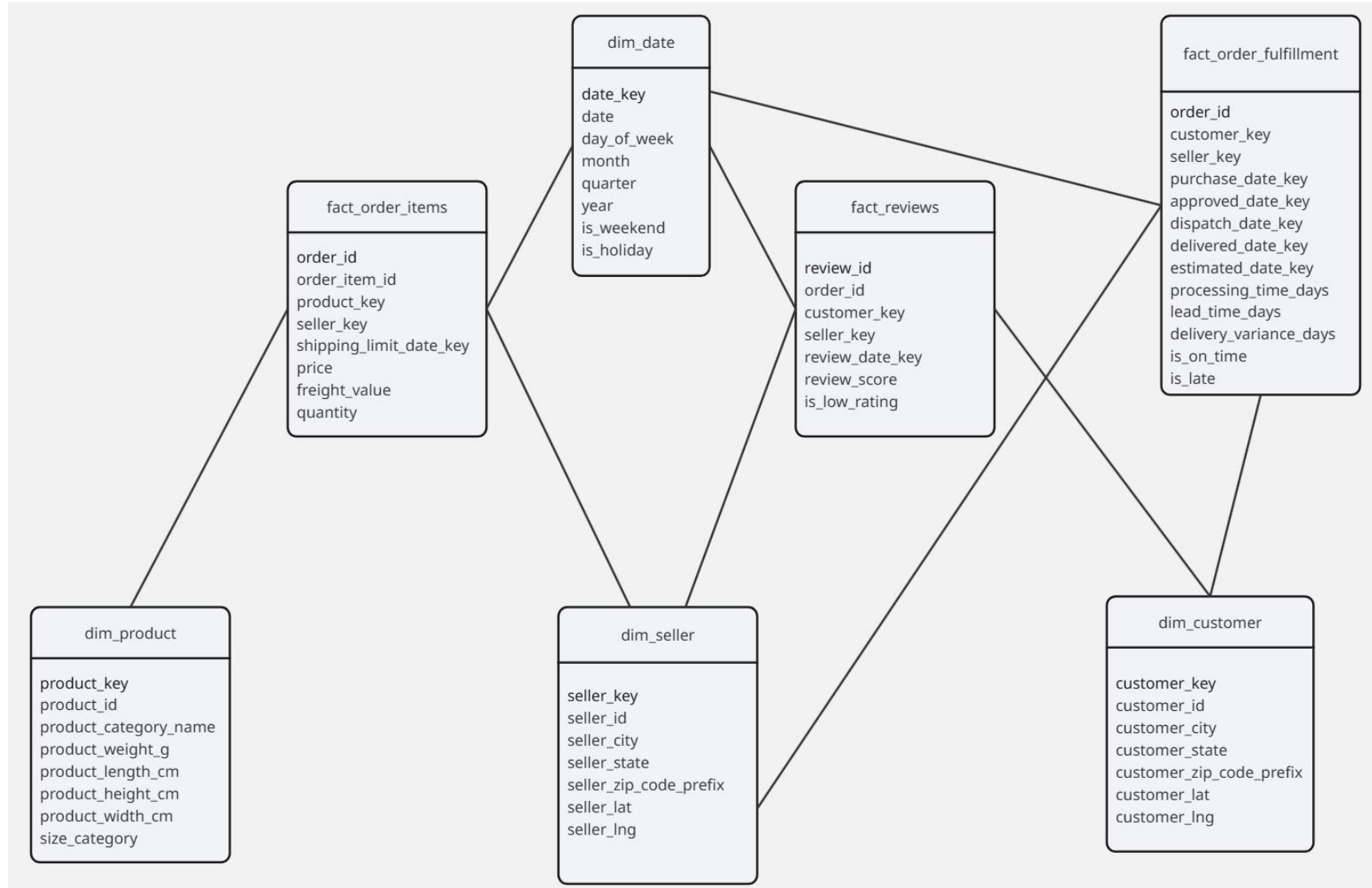
2) Referential Integrity Checks

- Verified foreign keys exist in the referenced tables.
- Used left anti-joins to return rows from one table that do not have a match in a second table.
 - E.g. Orders whose customer_id do not exist in the customers table

PySpark Performance Optimization

- DataFrame-Level Optimizations
 - Column Pruning
 - Broadcast joins for small lookups
 - Caching of reused DataFrames
- Query Execution Optimizations
 - Targeted Filtering Before Joins

Galaxy Schema



Multiple fact tables
sharing common
dimensions

Supports different
levels of granularity

Galaxy Schema – Pros and Cons

Advantages	Disadvantages / Challenges
Supports multiple business processes	More complex relationships and joins
Enables cross-domain analysis	Requires careful governance
Scalable and extensible	Potential query performance trade-offs

Pipeline Testing

Objective:

Verify data integrity and transformation accuracy throughout the pipeline.

Validation:

Random samples compared between layers to confirm transformation accuracy.

Expected Outcome:

Clean, structured data ready for Power BI visualization.

Functional Testing

Objective:

- Ensure the dashboards and data models work as expected.

Test Focus Areas:

- Data Connectivity
- Accuracy
- Filters & Slicers
- Navigation
- Performance

Testing – UAT

Objective:

- Validate that the dashboard meets business goals of improving delivery timeliness and customer satisfaction.

Key UAT Scenarios:

- Usability
- Relevance
- Insight Accuracy
- End-to-End Validation
- Business Impact

Probable Insights

Logistics

- On Time Delivery Percentage
- Delivery Variance
- Late Delivery Trend



Revenue

- Which product category has the most orders
- Which product category generated the highest revenue



Customer Satisfaction

- Product category with most Positive reviews
- Review rating proportion to total orders
- Customer feedbacks



Analysis & Visualization

olist Delivery Performance

Total Revenue

R\$ 15M

On Time Delivery

93.15%

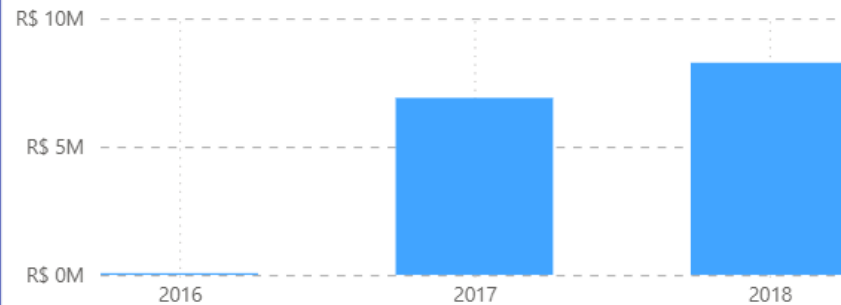
Median Delivery Variance

-12

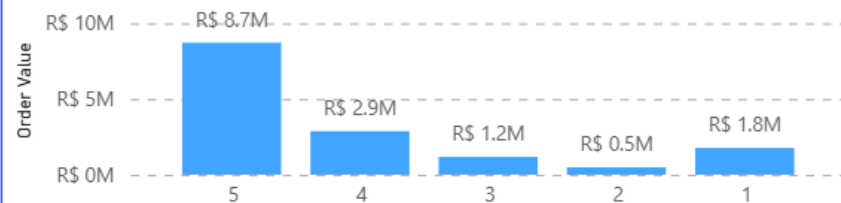
Year

All

Total Revenue

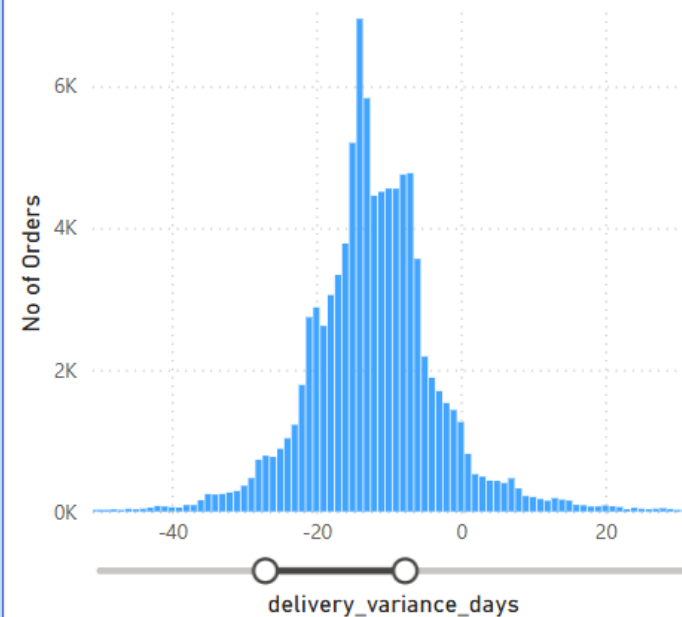


Order Reviews

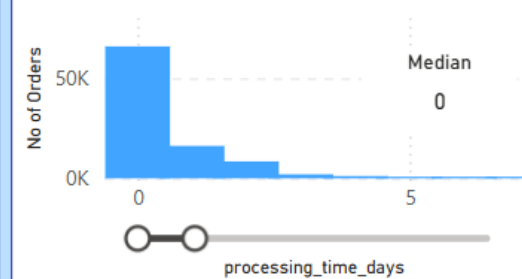


Review score	Total reviewed orders	Percent of total orders	Order Amount
5	55776	59.20%	R\$ 8,715,189
4	18587	19.73%	R\$ 2,875,803
3	7801	8.28%	R\$ 1,180,103
2	2869	3.05%	R\$ 488,389
1	9187	9.75%	R\$ 1,769,649
Total	94220	100.00%	R\$ 15,211,231

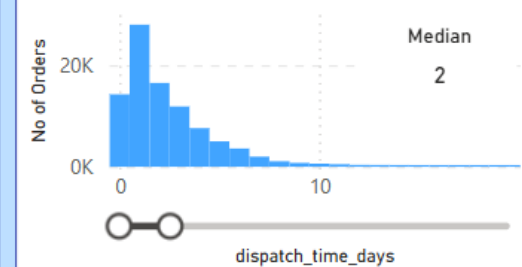
Delivery Variance(Days)



Order Possessing Time (Days)

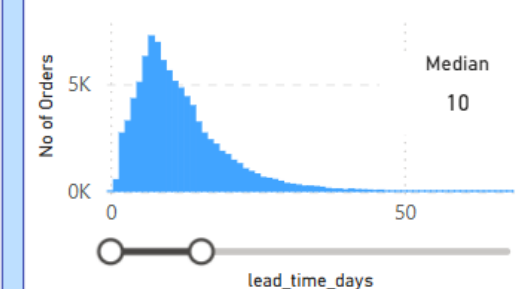


Order Dispatch Time (Days)



Product Category	Product Count	Category Contribution	Order Amount
health_beauty	9297	9.17%	R\$ 1,394,153
watches_gifts	5755	8.24%	R\$ 1,252,726
bed_bath_table	10775	8.08%	R\$ 1,229,213
sports_leisure	8331	7.32%	R\$ 1,114,006
computers_accessories	7546	6.74%	R\$ 1,025,601
furniture_decor	8053	5.90%	R\$ 898,203
Total	107024	100.00%	R\$ 15,211,231

Lead Time (Days)



Analysis & Visualization

olist Order / Rating

On Time Delivery

93.14%

Median Delivery Variance

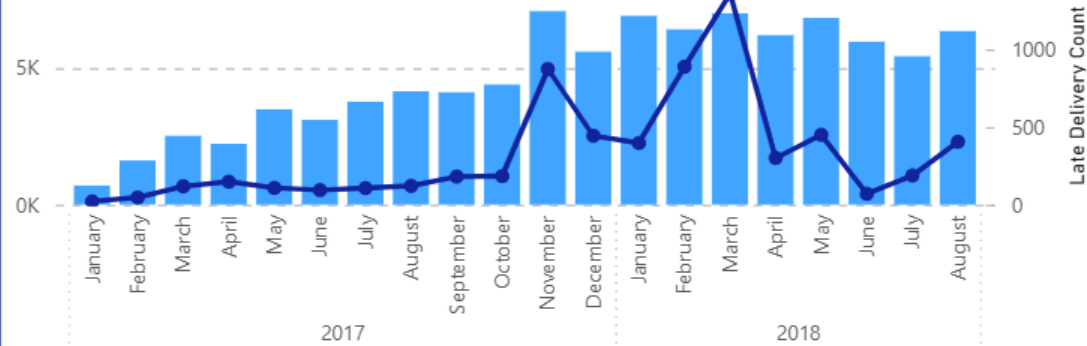
-12

Year

Multiple selec...▼

Order Trend

● Count of order_id ● No of late deliveries

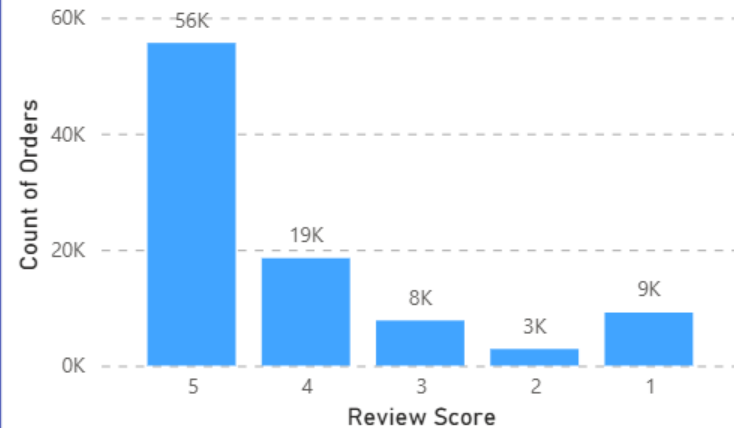


Review Comment

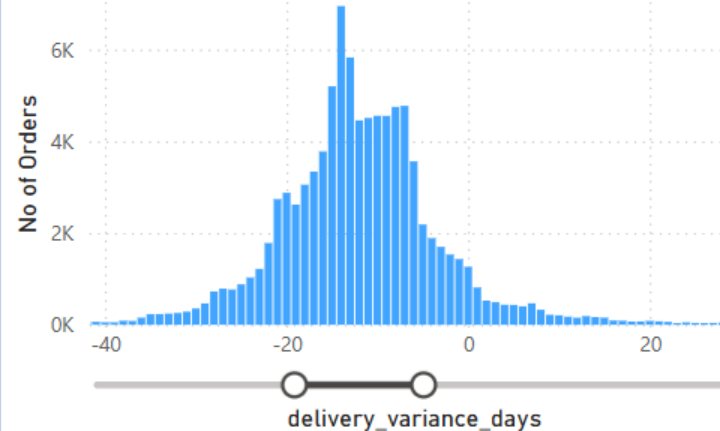
review_score

- 1
- 2
- 3
- 4
- 5

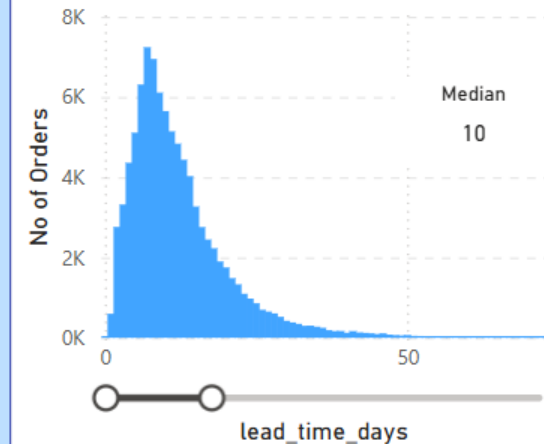
Order Reviews



Delivery Variance(Days)



Lead Time (Days)



Analytics and Insights

Insights



19.89% growth :
R\$8.27M



93.15%



Most Ordered :
Bed_Bath_Table



Most Profitable :
Health_Beauty



59.20%



9.75%



Wrong, defective, damaged Product
Incomplete, missing delivery
Slow, expensive shipping
Counterfeit product
No response

Business Recommendations

- Based on dashboard
 - Engage with logistics (3PL) partners to improve on delivery and handling to eliminate damages.
 - On board more SMBs with the right product categories to improve product category selection.
 - Engage and support SMBs that did not meet performance expectations.

Business Recommendations

- Based on pipeline
 - Improve data collection to increase accuracy and reliability of data
 - Increase feedback and communication with customers and businesses

Future Enhancements

- Utilize Great Expectations for automated validation at each layer
- Perform Integration Testing
- Build forecasting model in Power BI
- Fully automate pipeline

Thank You